

Certificate of Analysis

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Client:	New Brighton Bowling Club	Lab No:	4095199	shvbspv1
Address:	PO Box 18525 New Brighton Christchurch 8641	Date Received:	10-Feb-2026	
		Date Reported:	13-Feb-2026	
		Quote No:		
		Order No:		
Phone:	03 388 9059	Client Reference:		
		Submitted By:	Grant Hasson	

Sample Name: Green **Lab Number:** 4095199.1
Sample Type: TURF Cotula (S78)

Analysis	Level Found	Medium Range*	Low	Medium	High
pH	pH Units	5.9	5.8 - 6.5		
Olsen Phosphorus	mg/L	122	20 - 30		
Potassium	%BS	2.6	3.0 - 6.0		
Calcium	%BS	49	45 - 75		
Magnesium	%BS	8.9	5.0 - 15.0		
Sodium	%BS	1.9	0.0 - 5.0		
CEC	me/100g	17	12 - 25		
Total Base Saturation	%	63	40 - 80		
Volume Weight	g/mL	0.94	0.60 - 1.00		
K/Mg Ratio*		0.3	0.3 - 1.0		
Soil Sample Depth**	mm	0-75			
Soil Type**		Sedimentary			
me/100g		K 0.44 Ca 8.3 Mg 1.50 Na 0.32			
MAF Units		K 8 Ca 10 Mg 32 Na 14			

The above nutrient graph compares the levels found with reference interpretation levels. NOTE: It is important that the correct sample type be assigned, and that the recommended sampling procedure has been followed. R J Hill Laboratories Limited does not accept any responsibility for the resulting use of this information. IANZ Accreditation does not apply to comments and interpretations, i.e. the 'Range Levels' and subsequent graphs.

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Soil Analysis Results							
Sample Name:	Green						
Lab Number:	4095199.1						
Sample Type:	TURF Cotula						
Sample Type Code:	S78						
pH	pH Units	5.9	-	-	-	-	-
Olsen Phosphorus	mg/L	122	-	-	-	-	-
Potassium	me/100g	0.44	-	-	-	-	-
Potassium	%BS	2.6	-	-	-	-	-
Potassium	MAF units	8	-	-	-	-	-
Calcium	me/100g	8.3	-	-	-	-	-
Calcium	%BS	49	-	-	-	-	-
Calcium	MAF units	10	-	-	-	-	-
Magnesium	me/100g	1.50	-	-	-	-	-
Magnesium	%BS	8.9	-	-	-	-	-
Magnesium	MAF units	32	-	-	-	-	-
Sodium	me/100g	0.32	-	-	-	-	-
Sodium	%BS	1.9	-	-	-	-	-
Sodium	MAF units	14	-	-	-	-	-
CEC	me/100g	17	-	-	-	-	-
Total Base Saturation	%	63	-	-	-	-	-
Volume Weight	g/mL	0.94	-	-	-	-	-
K/Mg Ratio*		0.3	-	-	-	-	-
Soil Sample Depth**	mm	0-75	-	-	-	-	-
Soil Type**		Sedimentary	-	-	-	-	-

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Analyst's Comments

† Customer supplied data. Please note: Hill Labs cannot be held responsible for the validity of this customer supplied data, or any subsequent calculations that rely on this information.

Sample 1 Comment:

The medium or optimum range guidelines shown in the histogram report relate to sampling protocols as per Hill Laboratories' crop guides and are based on reference values where these are published. Results for samples collected to different depths than those described in the crop guide should be interpreted with caution. For pastoral soils, the medium ranges are specific for a 75mm sample depth, but if a 150mm sampling depth is used the nutrient levels measured may appear low against these ranges, as nutrients are typically more concentrated in the top of the soil profile. These soil profile differences are altered upon cultivation or contouring. Further explanation of the derivation of the medium and optimum ranges is available on request.

Summary of Methods

The following table(s) gives a brief description of the methods used to conduct the analyses for this job. The detection limits given below are those attainable in a relatively simple matrix. Detection limits may be higher for individual samples should insufficient sample be available, or if the matrix requires that dilutions be performed during analysis. A detection limit range indicates the lowest and highest detection limits in the associated suite of analytes. A full listing of compounds and detection limits are available from the laboratory upon request. Unless otherwise indicated, analyses were performed at Hill Labs, 28 Duke Street, Frankton, Hamilton 3204.

Sample Type: Soil

Test	Method Description	Default Detection Limit	Sample No
Sample Registration*	Samples were registered according to instructions received.	-	1
Soil Prep (Dry & Grind)*	Air dried at 35 - 40°C overnight (residual moisture typically 4%) and crushed to pass through a 2mm screen.	-	1
pH	1:2 (v/v) soil:water slurry followed by potentiometric determination of pH. In-house.	0.1 pH Units	1
Olsen Phosphorus	Olsen extraction followed by Molybdenum Blue colorimetry. In-house method.	1 mg/L	1
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	2 MAF units	1
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.01 me/100g	1
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.5 me/100g	1
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.04 me/100g	1
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.05 me/100g	1
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.1 %BS	1
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 %BS	1
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.2 %BS	1
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.1 %BS	1

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Sample Type: Soil			
Test	Method Description	Default Detection Limit	Sample No
CEC	Summation of extractable cations (K, Ca, Mg, Na) and extractable acidity. May be overestimated if soil contains high levels of soluble salts or carbonates. In-house.	2 me/100g	1
Total Base Saturation	Calculated from Extractable Cations and Cation Exchange Capacity.	5 %	1
Volume Weight	The weight/volume ratio of dried, ground soil. In-house.	0.01 g/mL	1

These samples were collected by yourselves (or your agent) and analysed as received at the laboratory.

Testing was completed between 11-Feb-2026 and 13-Feb-2026. For completion dates of individual analyses please contact the laboratory.

Samples are held at the laboratory after reporting for a length of time based on the stability of the samples and analytes being tested (considering any preservation used), and the storage space available. Once the storage period is completed, the samples are discarded unless otherwise agreed with the customer. Extended storage times may incur additional charges.

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Wendy Homewood
Operations Support - Agriculture